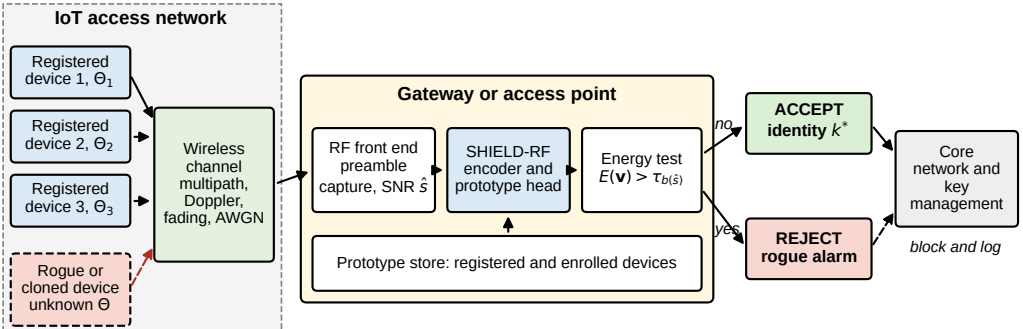
IoT access network
Registered device 1, Θ_1
Registered device 2, Θ_2
Registered device 3, Θ_3
Rogue or cloned device unknown Θ
Wireless channel multipath, Doppler, fading, AWGN
Gateway or access point
RF front end preamble capture, SNR $\hat{s}$
SHIELD-RF encoder and prototype head
Energy test $E(\mathbf{v}) > \tau_{b(\hat{s})}$
Prototype store: registered and enrolled devices
no
yes
ACCEPT identity k^*
REJECT rogue alarm
Core network and key management
block and log